



Enshrouded in nighttime darkness, however, Earth appears more as a black marble from space—one shimmering with light. The human search for light in darkness is long-standing. Many of our myths and religions focus on this search, yet it is only within the past century that humans have gained the ability to take flight—first to the skies and later into space—providing new vantage points from which to view Earth and the twinkling lights below, clearly visible at night.

For nearly 25 years, satellite images of Earth at night have served as a fundamental research tool, while also stoking public curiosity. These images paint an expansive and revealing picture, showing how humans have illuminated and shaped the planet in profound ways since the invention of the light bulb 140 years ago.

These lights and the darkness tell stories about our planet—stories that this volume will present for your consideration. *Earth at Night* will show how humans and natural phenomena light up the darkness, and how and why scientists have observed Earth's nightlights for more than four decades using both their own eyes and spaceborne instruments. It is an engaging and fascinating story; come along for the adventure!

[Above] NASA's *Black Marble*. These images of Earth at night were created in 2012 over 9 days in April and 13 days in October. It took the Suomi National Polar-orbiting Partnership (NPP) satellite 312 orbits and 2.5 terabytes of data to get a clear shot of every parcel of Earth's land surface and islands. These new data were mapped over existing Blue Marble imagery to provide a realistic nighttime view of the planet. (See "Making a Cloud-Free, Global, Earth-at-Night Image Using NASA's Black Marble Product Suite" in *Appendix B* for technical details.)

Vision and Remote Sensing

Seeing Is Sensing

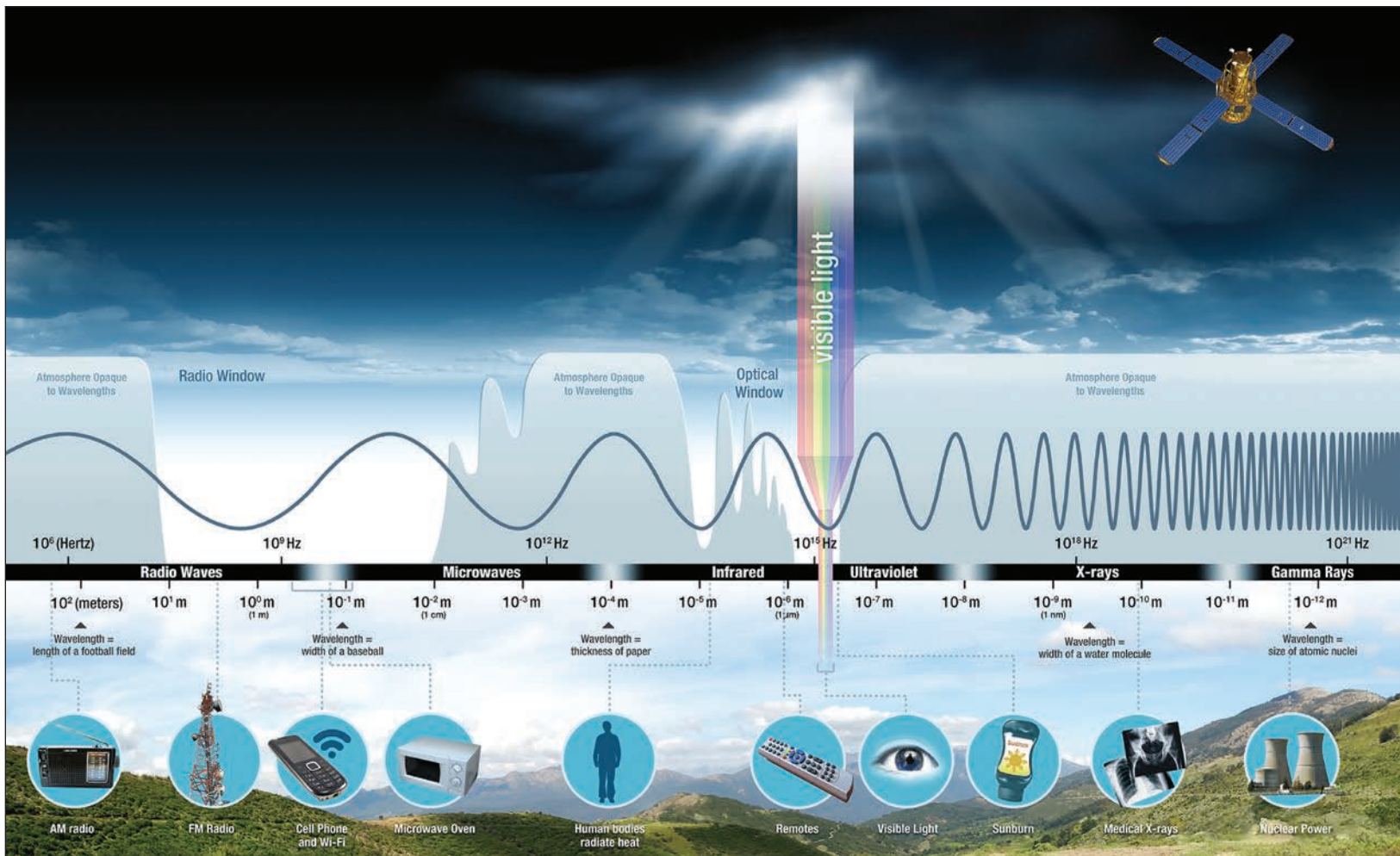
Seeing, or vision, is one of several senses we possess. When you see something, your visual system absorbs light—photons—and your brain processes the view into something you can understand and, perhaps, act upon.

Humans have long extended their visual capabilities with instruments—such as telescopes—to enable them to see across distances that their own eyes could never possibly see. This is a form of “remote sensing”—a capability that is at the core of our new understanding of how Earth “works.”

Over time, science and technology developments have enabled us to create more-sensitive observing tools that far exceed the capabilities of relatively simple telescopes, providing the ability to see things we had never even dreamed of previously. Our technologies can record phenomena well outside the realm of human visual abilities—things like heat and moisture, the measurement of which would have seemed magical to our forebears. But when it comes to inspiring the human spirit, there’s nothing quite like visible light, so light at night will be our focus here.

What Is Light?

Unless an object has a temperature of absolute zero (-273°C) it reflects, absorbs, and emits energy—called electromagnetic radiation—in ways that depend on its physical and chemical properties. The amount of electromagnetic radiation an object emits depends primarily on its temperature. The higher the temperature of an object, the faster its electrons vibrate and the shorter the peak wavelength of the emitted



radiation. Conversely, the lower the temperature of an object, the slower its electrons vibrate, and the longer its peak wavelength of emitted radiation. Given the wide range of temperatures in the universe, it should be clear that electromagnetic radiation has wavelengths that span a very wide range.

The electromagnetic spectrum is the range of traveling waves of energy that run from very short gamma and X-rays through ultraviolet light, visible light, microwaves, and out to long radio waves. Visible light is the quite narrow band that human eyes are adapted to see. But as noted earlier, humans are proficient at developing tools to extend the range of their capabilities. Using sensors on orbiting satellites, designed to detect multiple spectral-band combinations, scientists can “tune in” to study various aspects of Earth’s surface in ways not possible from a simple color photograph. Just as soils,

[Above] *Electromagnetic spectrum*. The electromagnetic spectrum is the range of energy waves of various wavelengths—from the very short gamma and X-rays [right], through longer ultraviolet, visible, and infrared light, to microwaves and very long radio waves [left].

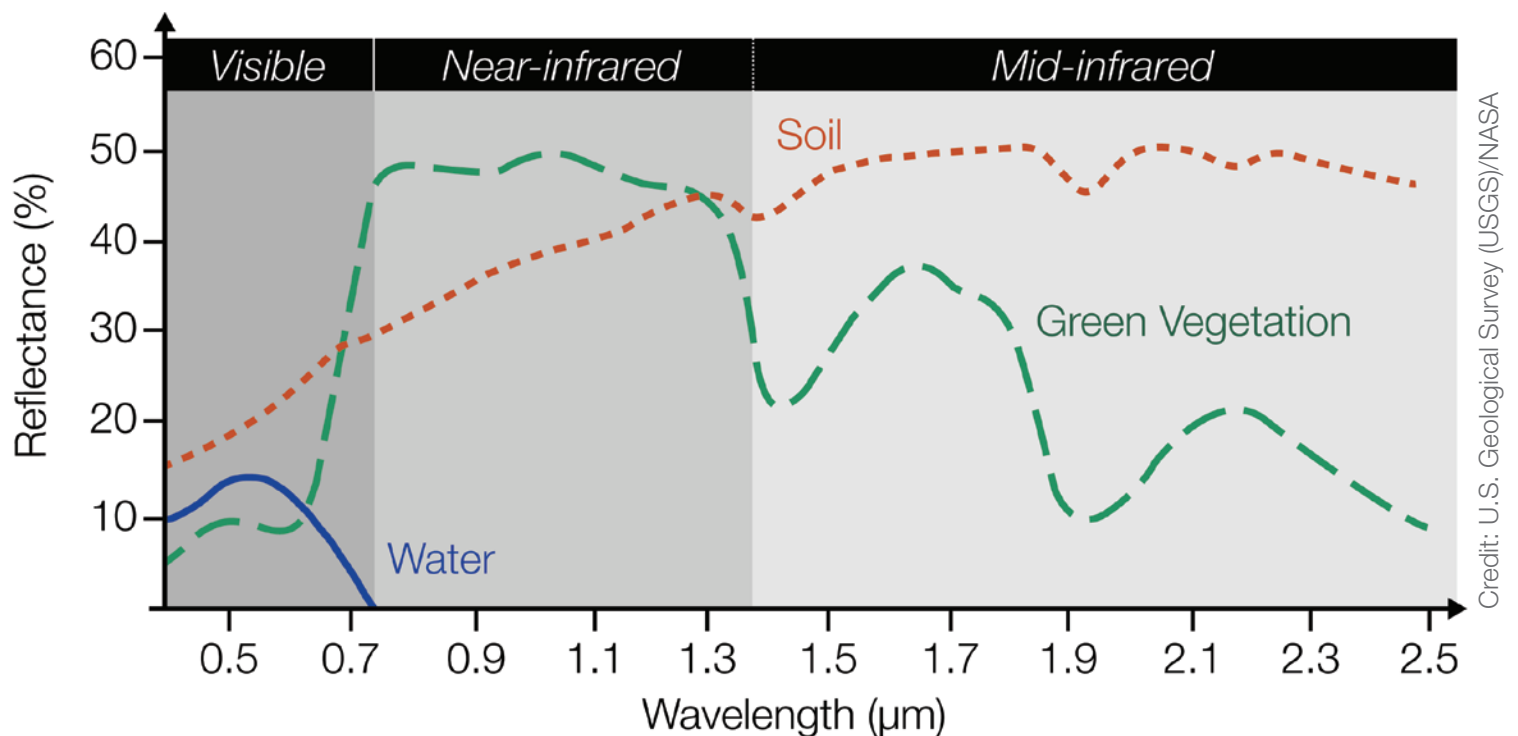
different plant types, plant health, the presence of water, bare rock, ice, and many other types of land cover each have unique “signatures” in the electromagnetic spectrum, different sources of light (e.g., incandescent lamps and LEDs) have their own unique signatures in the electromagnetic spectrum. Over time, scientists can observe and analyze changes in spectral signatures to detect changes in Earth’s surface and identify various phenomena taking place (including those dealing with water, vegetation, soil, and the like—see the “Spectral signatures” figure below), sometimes based on lighting patterns.

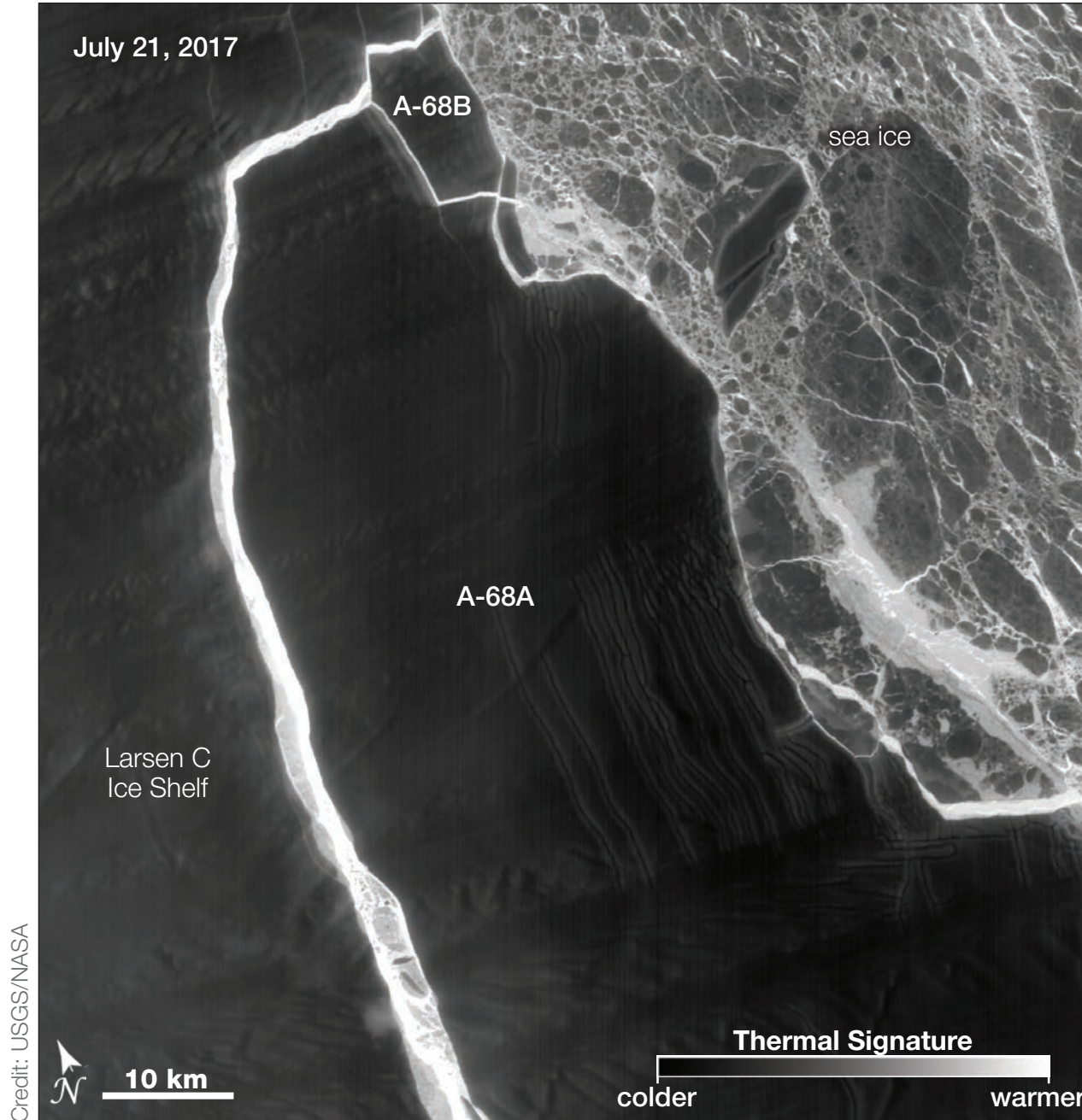
Remote Sensing

[Below] *Spectral signatures.* Satellite instruments measure light emitted or reflected back to space at different wavelengths and create *spectral reflectance curves*. Differences in the shape of the “spectra” can be used to determine what the satellite is “seeing,” such as soil, green vegetation, and water.

Placing vision-augmenting tools—sensors—on satellites that orbit Earth has expanded our ability to see our home planet in a new light.

Observing Earth from space using sensors on satellites (i.e., remote sensing) traces its origins to the early days of the space age—both Russian and American programs—when surface-imaging sensors first flew onboard aircraft and then later on spacecraft. With the emergence of manned space flight in the 1960s, Earth-orbiting cosmonauts and astronauts acted much like tourists by taking photos from the windows of their spacecraft.





[Left] *Seeing ice at night.* A massive iceberg (A-68) first broke away from Antarctica's Larsen C ice shelf sometime between July 10 and July 12, 2017. The fractured berg (A-68A and A-68B) and shelf are visible in this image, acquired on July 21, 2017, by the Thermal Infrared Sensor (TIRS) on the Landsat 8 satellite. This false-color view shows the relative warmth or coolness across the region. White indicates where the ice or water surface is warmest; dark grays and blacks are the coldest areas of ice.

Remote sensing is more than simply taking pictures, however. It more fully describes the science—and art—of observing, identifying, and measuring an object without coming into direct contact with it. This process involves detecting and measuring radiation of different wavelengths emitted or reflected from distant objects or materials. Emitted light is comprised of photons in an excited state; for example, light coming from the Sun or a light bulb filament. Reflected light is light from another source bounced off an object's surface.

Orbiting Tools to Observe Nightlights



The original motivation for making low-light Earth observations was practical: military meteorologists wanted to track clouds and storms, smoke plumes, and dust storms at all hours, to better support military planning and execution. The Air Force, in particular, needed to provide more accurate and timely forecasts for pilots working at night over both land and sea.



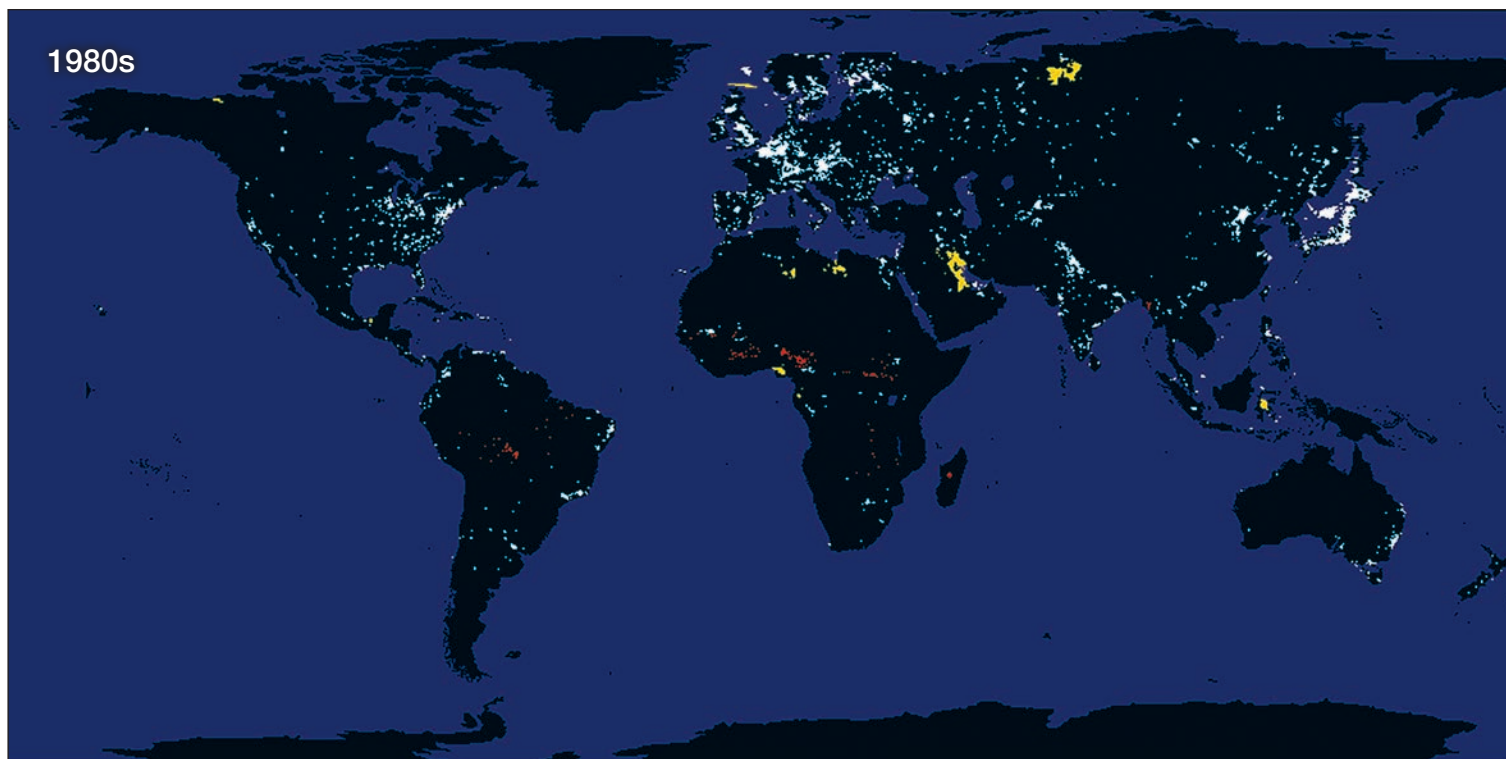
[Previous pages] *Vintage photos of Earth at night—NASA’s Mercury-Atlas mission*. The photograph on page 10 was taken from NASA’s unmanned Mercury-Atlas 4 mission on September 13, 1961. High clouds (white) are mixed with Earth’s nightlights (yellow) on the right-hand side of the image, with Earth’s horizon on the left-hand side. The handwritten annotation at the top (that appears backwards in that photo) indicates this was the 348th photograph taken on the MA-4 mission. The next photograph (page 11) taken by MA-4, number 349, depicts an even darker scene of Earth at night. The white line in this photo is due to a film artifact.

So, scientists have observed Earth’s nightlights for more than four decades, first with astronaut photography and military satellites. Since the 1960s, the U.S. Air Force has operated the Defense Meteorological Satellite Program (DMSP), a series of 18 polar-orbiting satellites that observe clouds and other weather variables in key wavelengths of infrared and visible light. Starting in 1972, the DMSP satellites included the Operational Linescan System (OLS), which gives weather forecasters some ability to see in the dark.

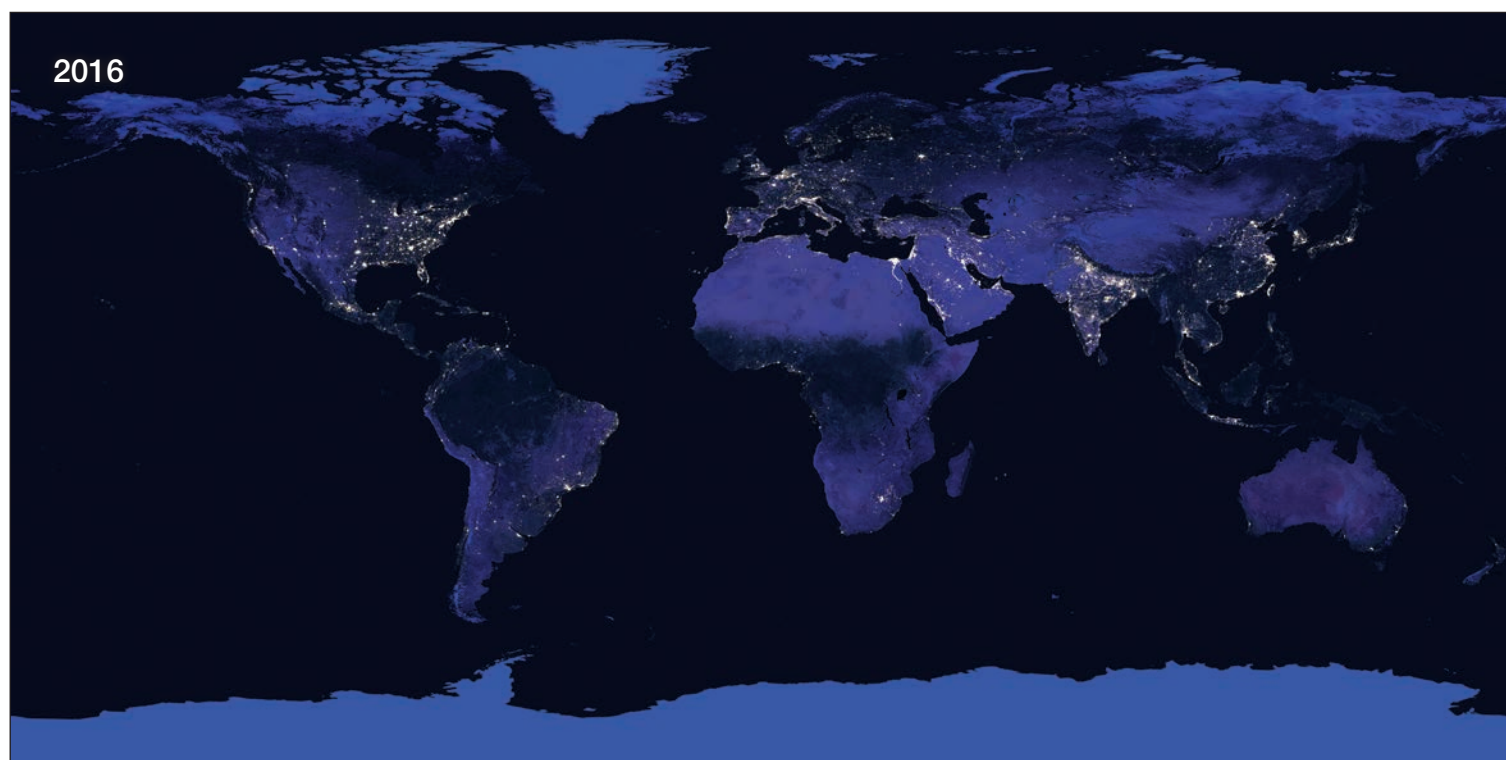
While DMSP has been a source of nighttime images for decades, until fairly recently the data were classified, which meant that only a few civilian scientists could conveniently gain access to study the data. The atmospheric science community was eager to have a more accessible night-vision tool to better understand weather and climate patterns and phenomena. Finally, in 2011, a new source of unclassified satellite images of Earth at night became available—one that improved upon the capabilities of OLS. The new low-light sensor was called the Visible Infrared Imaging Radiometer Suite (VIIRS), and was launched in October 2011 onboard the Suomi National Polar-orbiting Partnership (NPP) satellite—a partnership between NASA, the National Oceanic and Atmospheric Administration (NOAA), and the U.S. Department of Defense.

VIIRS currently flies on Suomi NPP as well as on the NOAA-20 satellite (launched November 2017). The VIIRS “Day/Night Band,” or DNB, can observe dim light down to the scale of an isolated highway lamp or fishing boat. It can even detect faint, nocturnal atmospheric light—known as airglow—and observe clouds lit by it. Through the use of its DNB, VIIRS has made the first quantitative measurements of light emissions and reflections from Earth’s surface, distinguishing the intensity and the sources of nighttime light. The combination of these measurements gives us a global view of the human and natural light sources on Earth. Looking ahead, three more satellite missions are planned over the next decade, with a VIIRS instrument scheduled to be onboard each platform.

Thanks to advancements in sensor technology and improved optics, the VIIRS DNB is ten-to-fifteen times better than the OLS sensor at resolving the relatively dim

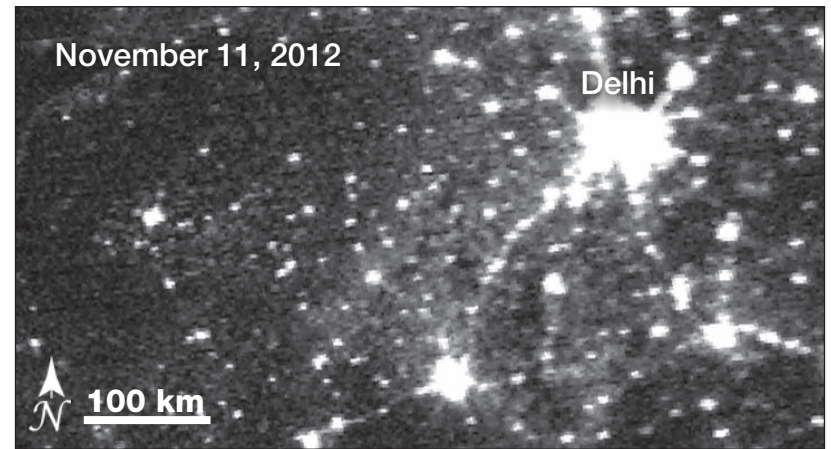


[Above] *Early map of Earth at night.* Technology has come a long way since the 1980s and 1990s. This map shows the visible light sources on Earth's surface as measured by the OLS on the DMSP satellites during the 1980s. White dots represent city lights, red dots represent forest fires, and yellow dots represent oil-well fires.

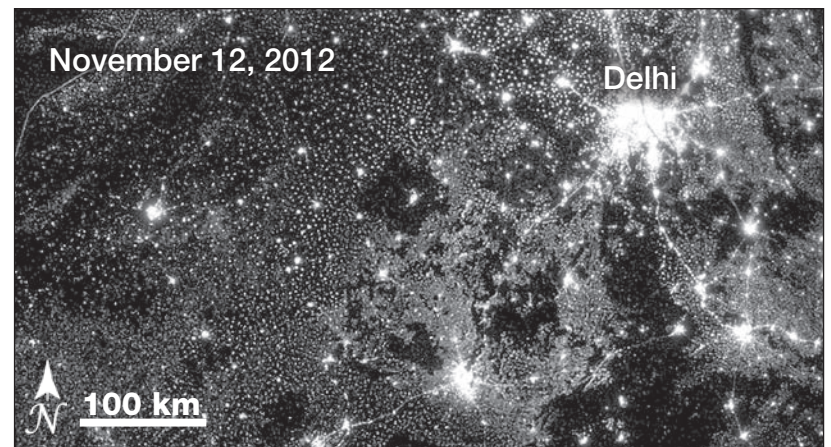


[Above] *The 2016 Black Marble.* The 2016 Earth at night image, dubbed NASA's *Black Marble*, was created with VIIRS DNB data from the Suomi NPP satellite. The data were acquired over 190 days in 2016. This translates into 5247 orbits and 42 terabytes of input data to get a cloud-free composite image of Earth's land surfaces and ocean.

[Right] This image pair illustrates the differences in resolution between OLS and VIIRS. On November 11, 2012, the OLS on a DMSP spacecraft captured the top image of city, village, and highway lights near Delhi, India. For comparison, the VIIRS instrument on the Suomi NPP satellite acquired the bottom image showing this same area one night later.



OLS



VIIRS

lights of human settlements and reflected moonlight. Just like the picture on your television set, satellite images are made up of tiny squares. These squares are called pixels—short for picture elements—and represent the amount of reflected or emitted light energy recorded for that part of the image. Each VIIRS pixel covers a distance of roughly 0.46 miles (742 meters) across, compared to the 1.86-mile (3-kilometer) footprint of OLS. To learn more about the technical details of OLS and VIIRS, see *Appendix C*.

As they have done since the dawn of the space age, astronauts onboard the International Space Station (ISS) also take photographs of Earth at night. In fact, their images have even higher spatial resolution than images from VIIRS (typically tens-to-hundreds of meters per pixel). However, unlike satellites, astronauts are not looking at Earth 24 hours a day, 7 days a week, 365 days a year. Due to the orbit of the ISS, astronauts